Name _____ Period _____ Date _____

2-6 Lesson Quiz: The Quadratic Formula

short form

1. Solve $x^2+8x-6=0$ using the Quadratic Formula.

Ⓐ $x = 32 + 2\sqrt{2}$ and $x = 32 - 2\sqrt{2}$

Ⓑ $x = -4 + \sqrt{22}$ and $x = -4 - \sqrt{22}$

Ⓒ $x = 12$ and $x = -\frac{1}{2}$

Ⓓ $x = 4 + 2\sqrt{22}$ and $x = 4 - 2\sqrt{22}$

Show your work: $a =$ ____, $b =$ ____, $c =$ ____

2. Solve $x^2+2x+7=0$ using the Quadratic Formula. Select **all** solutions that apply.

☐ **A.** $x = -1 + i\sqrt{6}$

☐ **B.** $x = -1 + \sqrt{6}$

☐ **C.** $x = -1 - i\sqrt{6}$

☐ **D.** $x = -1 - \sqrt{6}$

☐ **E.** $x = 1 - \sqrt{6}$

Show your work:

3. Describe the nature of the roots for the equation $49x^2 - 28x + 4 = 0$.

Ⓐ two real roots

Ⓑ one real root

Ⓒ two complex roots

Ⓓ one complex root

Show the discriminant: $b^2 - 4ac =$

Explain your choice in one sentence using the word *discriminant*:

Scoring: Question 1 and Question 3 are 10 points each. Question 2 is a bonus of up to 5 points (both correct boxes, none wrong). Total out of 20.

2-6 Lesson Quiz, short form: answer key

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1. **B, $x = -4 \pm \sqrt{22}$.** $a = 1$, $b = 8$, $c = -6$: $x = \dfrac{-8 \pm \sqrt{64 + 24}}{2} = \dfrac{-8 \pm \sqrt{88}}{2} = \dfrac{-8 \pm 2\sqrt{22}}{2} = -4 \pm \sqrt{22}$. Savvas quiz item 1, DOK 2, Realize skills A31, A72 (Example 1A). **10 points:** 6 for B, 4 for a, b, c and a correct substitution.

2. **A and C, $x = -1 \pm i\sqrt{6}$.** $a = 1$, $b = 2$, $c = 7$: $x = \dfrac{-2 \pm \sqrt{4 - 28}}{2} = \dfrac{-2 \pm \sqrt{-24}}{2} = \dfrac{-2 \pm 2i\sqrt{6}}{2} = -1 \pm i\sqrt{6}$. B and D drop the i (the radicand -24 is negative); E has the wrong sign on -1 and also drops the i. Savvas quiz item 2, DOK 2, Realize skills A31, A72 (Example 1B). **Bonus 5:** A and C and no other box; 3 for one of A or C and no wrong box; 0 otherwise.

3. **B, one real root.** $b^2 - 4ac = (-28)^2 - 4(49)(4) = 784 - 784 = 0$, so there is one real root ($x = \frac{28}{98} = \frac{2}{7}$; the equation is $(7x - 2)^2 = 0$). Savvas quiz item 3, DOK 1, Realize skill A32 (Example 3). **10 points:** 6 for B, 2 for the discriminant 0, 2 for the sentence naming the discriminant.

Items left off this form (available for a retake or Form B): Savvas item 4, Richard's ball, $h(t) = -5t^2 + 10t + 6$: does it reach 12 m? (answer D, the ball does not reach 12 m: $-5t^2 + 10t - 6 = 0$ has discriminant $100 - 120 = -20 < 0$; DOK 2, A31, A72; taught by Example 4, cut, and rehearsed by the packet's Practice page); Savvas item 5, values of b that give $27x^2 + bx + 3 = 0$ one real solution (answer A and F, $b = \pm 18$: $b^2 - 4(27)(3) = b^2 - 324 = 0$; DOK 2, A32; taught by Example 5, cut).

IXL replacement: SmartScore 70 or above on YQH and QHK replaces this quiz with 20/20.